



OPEN Household vulnerability to climate impacts on WASH and risk mitigation in informal settlements of Eastern Uganda

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Climate-related hazards such as floods and droughts disrupt household access to safe water, sanitation, and hygiene (WASH), undermining progress toward the Sustainable Development Goals (SDGs). Evidence on how household characteristics such as housing type, altitude, subjective social status, and disability status shape vulnerability to floods and drought and related risk mitigation in informal settlements remains limited. This study, therefore, assessed the vulnerability to climate impacts on WASH, risk mitigation, and its determinants among households in informal settlements in Eastern Uganda. This cross-sectional study applied nexus thinking and used an adapted questionnaire, informed by the World Health Organisation's climate vulnerability checklist, to collect data from 589 households in Eastern Uganda. Quantitative data on household characteristics, documented flood and drought impacts, as well as risk mitigation measures, were collected. Robust least squares regression was used to show predictors of climate impact scores as well as risk mitigation scores at 95% confidence level. Results show that 21.4% reported increased susceptibility to waterborne diseases due to contamination from animal faeces and sewage. Vulnerability to flood impacts was higher among households living in temporary housing or with disabled members, but lower among those in higher-altitude areas or with higher subjective socioeconomic status. Higher subjective social status was also linked to reduced drought impacts, whereas temporary housing reduced the capacity to mitigate drought risk. Overall, climate-related hazards compromise WASH in informal settlements, with cascading effects on health, education, food security, gender equality, and poverty reduction. These findings highlight the need to strengthen water governance, promote integrated (nexus) approaches, and build resilient WASH systems through multi-sectoral partnerships to mitigate climate risks in vulnerable communities.

Keywords WASH, Water governance, Informal settlements, Nexus, Vulnerability, Resilience, Risk mitigation, Uganda

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Abbreviations

SDGs	Sustainable development goals
SSS	Scale of subjective social status
UBOS	Uganda Bureau of Statistics
UN	United Nations
WASH	Water, sanitation, and hygiene

Globally, more than 3 billion people live in settings prone to climate-related hazards, and 700 million are at risk of being displaced by drought alone before the end of the SDG era in 2030¹. Climate-related hazards, such as storms, floods, droughts, and extreme temperatures, have become a norm in cities in low- and middle-income countries^{2–6}, although they have also been reported in the developed world⁷. More than 200,000 Ugandans are affected by climate-related hazards annually⁸. As of July 2021, 26,000 people in Uganda had been displaced by climate-related hazards⁹. In July 2022, about 6,630 people in the Elgon region, where Mbale City, our study area, is located, were displaced by flash floods, mudslides, and landslides⁹.

Mbale city faces recurrent floods, flash floods, prolonged droughts, and landslides, all intensified by climate change and increased vulnerability due to its location in a mountainous and low-lying zone, such as the Nabuyonga valley, which makes it a disaster-prone area¹⁰. The July 2022 floods affected approximately 18,000 people, destroyed more than 7,000 farms, displaced over 5600 people, and cut off the clean water supply for about 400,000 people^{11,12}. Additionally, the increased frequency of mudslides on the Elgon slopes has killed dozens in recent years, due to the changing rainfall patterns^{10,13,14}. Increased frequency of mudslides on the Elgon slopes around Mbale has killed dozens in recent years, with the events linked to changing rainfall patterns and deforestation. Climate-related hazards, including drought, are reported to cause a significant loss of property and other sources of livelihood in Mbale City^{8,10,13,15}. Therefore, both the forcibly displaced persons and host communities facing comparable vulnerabilities end up accommodated in informal settlements surrounding the city¹⁶.

Distinct vulnerabilities to climate risks characterize informal settlements due to inherent challenges and infrastructural deficiencies¹⁷. Patterns of urbanisation, urban planning, and land use further influence the vulnerability to climate-related hazards in informal settlements¹⁸. Residents in these settlements also lack access to many resources that help protect against climate hazards. Precarious and poor-quality housing, for instance, offers no protection against flooding or extreme temperatures. So, the effect on people's health and well-being is greater than if they were living in households with at least adequate insulation and ventilation. With livelihoods compromised and a lack of social protection or insurance, the impacts of climate disasters may keep families in poverty for generations¹⁹. Although not widely studied in the context of informal settlements in cities in the global south, climate-related hazards lead to the destruction of water, sanitation, and hygiene (WASH) infrastructure, including sanitation facilities and water pipes, disrupted services, and inhibited system efficacy, including a deterioration of water quality^{20–23}.

Climate-related hazards worsen the already poor state of WASH services in informal settlements^{24,25}. To remain resilient and to cope with poverty and the effects of climate-related hazards, rainwater harvesting has been an option for households in informal settlements in the global south²⁶ and elsewhere^{27,28}. However, in the case of East Africa, the dry spells are becoming longer than ever reported in the last 40 years²⁹, which compromises resilience. The negative impacts of climate-related hazards are aggravated by a lack of infrastructure and institutions to foster resilience¹⁸. The negative impacts of climate-related hazards on WASH (SDG 6) have negative implications for the attainment of other SDGs in cities^{18,30,31}. Access to safe WASH services is a human right, whose inadequacy contributes to the spread of communicable diseases such as cholera and hepatitis E^{32,33}, thereby hindering the attainment of good health and well-being (SDG 3). Climate impacts on WASH services are associated with public health risks, such as diarrhoea, which contributes to malnutrition (SDG 2)³³ and compromises inclusiveness, safety, resilience, and sustainability in cities (SDG 11)^{34,35}. The interaction between WASH and other SDGs is crucial in alleviating poverty among the urban poor; for example by reducing healthcare costs, preventing loss of income due to illness, and safeguarding livelihoods (SDG 1). Furthermore, it's central to upholding gender equality (SDG 5) by lessening the disproportionate burden on women and girls in collecting water and managing sanitation, which enables them to participate more fully in education, work, and community life^{36,37}.

Addressing WASH through the climate-related hazards lens requires a multi-sectoral and collaborative approach²¹, where stakeholders directly or indirectly involved in implementing other SDGs are brought on board. However, to achieve that, it is pertinent to generate evidence of how climate-related hazards affect WASH and the implications for other SDGs. In 2010, Uganda developed a national policy for disaster preparedness and management to strengthen prediction, mitigation, preparedness, response, and recovery to disasters³⁸. The implementation of the national policy to foster resilience and adaptation is hampered by limited funds, including for contingency plans, and the lack of a national law on disaster risk reduction and management³⁹. This study, therefore, assessed the household vulnerability to climate impacts on WASH and risk mitigation in informal settlements of Eastern Uganda. It further employed a nexus thinking⁴⁰ to illustrate the intricate relationships between climate (SDG 13) impacts on WASH and the attainment of other SDGs.

Materials and methods

Study design and setting

We applied a cross-sectional study design to collect data from informal settlements in Mbale City, the main municipal, administrative, and commercial centre of Mbale and surrounding districts in elgon region of eastern Uganda districts. Mbale City is encircled by Mbale District, which borders the districts of Tororo to the south, Manafwa to the southeast, Sironko to the northeast, Kumi to the north, Budaka to the northwest, and Butaleja.

It covers an area of 2,467 square kilometres (953 sq mi). Mbale city is located at the foot of the Wanale ridge of Mt. Elgon, at an average elevation of 1,156 m (3,793 ft) above sea level and a land area of 2,435 hectares. Mbale City has a population of 290,414, among which 13,319 are above the age of 60 years, and 622 are refugees⁴¹. The city has 74,291 households, averaging a size of 3.9; 17.8% engage in a subsistence economy, and 53.5% are food insecure⁴¹. The city has previously experienced climate-related hazards, such as floods and droughts, and associated impacts, including a high incidence of vector-borne diseases, including malaria⁴². Mbale was selected as the study site because it exhibits geographical, socio-economic, and infrastructural characteristics that are representative of many secondary cities in Uganda and across sub-Saharan Africa. The city previously experienced outbreaks of waterborne diseases, such as diarrhoea, cholera, and typhoid^{43–45}. Mbale experiences typical problems associated with urban areas, including high population growth and unemployment, lack of basic urban services, and inadequate WASH services^{43,46}. Only 59.9% of the households in Mbale City use improved sanitation facilities, and 93.6% have access to an improved water source⁴¹ (Fig. 1).

Sample size and sampling

We estimated the sample size using the Kish-Leslie formula for cross-sectional studies⁴⁷. During sample size estimation, we assumed a conservative proportion of households experiencing WASH impacts due to climate-related hazards of 50%, based on limited evidence in the study domain. We used a 95% level of confidence, a margin of error of 0.05, and a Z score of 1.96. Factoring in a design effect of 2.0 due to intra-cluster correlation⁴⁸ And a non-response rate of 10% yielded a minimum sample size of 535 respondents. We considered a non-response rate of 10%, as individuals in urban settings, such as Mbale City, often have limited time to participate in surveys^{49,50}. We also increased the sample size from 535 to 589 to improve statistical power and precision^{51–53}. The research team applied a multi-stage sampling technique to select respondents. Firstly, the team purposively selected the Northern and Industrial Divisions for geographical representation of the city. Secondly, the team randomly selected 50% of the wards or parishes in each division. The third stage involved using a list of all cells and villages for the selected wards, obtained from the area's local council chairpersons, to randomly select 10% of the cells and villages. With the help of a list obtained from the respective local council chairpersons, we used the Microsoft Excel random number generator to select households from 10% of the selected cells or villages. At the household level, we purposively selected the household head or the second-in-command to respond to the survey tool. The research assistants purposively selected and interviewed household heads or the second-in-command, as they had the primary responsibility of ensuring access to WASH services during climate events^{54,55}.

Measurement of study variables

Dependent variables

The dependent variables included scores for the impacts of flood and drought on WASH services, as well as the respective risk management scores. Operationally, we defined floods as the overflow of water onto normally dry land caused by heavy rainfall, poor drainage, or blocked waterways, resulting in temporary disruption of living conditions and access to WASH services. We operationally defined drought as a prolonged period of little or no rainfall leading to water scarcity and reduced access to safe water and sanitation. To assess the impacts of floods and droughts on WASH, we asked household heads if they had experienced floods or droughts in the past 12 months. Household heads that had experienced climate-related hazards were asked how these impacts affected WASH services and the risk mitigation measures they had in place. We defined risk management strategies as



Fig. 1. Map showing the location of Mbale City and the surrounding districts and in relation to Uganda. (Created using Qgis).

the practical actions taken by households to prepare for, reduce, or cope with the impacts of floods and droughts on WASH services within the household.

Scoring approach for flood impacts and risk management

To assess the impacts of floods on WASH, we based our scores on respondents' experiences. We asked whether they encountered the following situations: disruption of wastewater or sewage systems, cross-contamination from damages to sewage systems, increased susceptibility to water borne diseases due to animal faeces and sewage washing into surface water (water contamination), decreased capacity to provide or access safe drinking and cooking water, inability to access or provide hygiene services, damage to waste storage leading to environmental contamination by biological and chemical hazards, decreased capacity to maintain waste collection and treatment, damage to emergency or alternative water sources, reduced ability to use toilets, showers, damage to hygiene facilities like tippy taps, temporary water supply interruptions, decreased ability to use laundry services including wire lines, and potential rodent infestation around rubbish bins. Each experience was assigned a point value (0 or 1), with a maximum possible value of 13 points.

To establish the risk management scores for floods, we asked respondents if they had: a natural floodwater infiltration system to reduce risk of flooding, an anti-mosquito breeding measures, a schedule for emptying latrines in advance of the flood season to avoid overflows, a safe waste storage place, a safe waste disposal system before, during and after floods, an established safe management approach to transport waste in case of floods, water storage tanks appropriately covered to prevent access or contamination, and safety located for flooding events, an on-site water purification equipment to provide safe drinking water, waste pits able to withstand flood events, or kept waste sealed in rubbish bins to avoid rodents. Each flood risk management measure was assigned a point value (1 or 0), with a maximum of 10 points. The authors and experts guided on the scoring approach; opting for uniform score for each flood impact and risk management strategy. The research team noted that there were no standardized approaches currently that guide on quantification these impacts.

Scoring approach for drought impacts and risk management

The impact scoring approach used was based on respondents' experiences of the effects across different domains. We asked respondents if they had experienced the following disruptions as a result of drought: limited availability of water for domestic use, increased water salinity in groundwater resources due to decreased recharge, limited access to potable water for drinking and cooking, poor water quality, increased rate of broken pipes, reduced function of sanitation systems and hygiene practices (flush toilets, showers, sewerage, hand washing), increased dependence on less secure alternative water sources, and increase in vector breeding sites due to inadequate water storage in the household or surrounding areas. Each drought impact on WASH was assigned a single point value (otherwise 0), and all effects had equal weights as guided by a team of climate change and health experts. The maximum score was therefore eight points.

To establish the risk management scores for droughts, we asked respondents if they had: a water management system that reduces vector breeding sites, anti-mosquito breeding measures, a rainwater catchment system with safe water storage, water storage tanks with appropriate covers to prevent contamination, water storage that is protected from direct sunlight and sufficient drinking water for visitors. Each drought risk management measure was assigned a point value, with a maximum of six points; similar approach that was used on flood impacts and mitigation.

Independent variables

The independent variables included household background characteristics such as age, sex, level of education, employment status, occupation, housing type, and subjective social status. The data collection tools also captured data on altitude since elevation

affects vulnerability to climate-related hazards and household resilience^{56–59}. We classified the type of housing as (1) permanent for houses made of brick with iron sheets/tiled roof, (2) semi-permanent for houses made of mud and wattle and roofed with iron sheets, or temporary for houses made of mud and wattle and roofed with tarpaulin or grass.

We generated scores for subjective social status using the MacArthur Scale of Subjective Social Status (MacArthur SSS Scale)^{60–62}. The MacArthur SSS Scale has been used since 2000 to assess how individuals perceive their social status relative to others in their neighbourhood⁶². The ladder contains ten rungs from which individuals choose where they feel they belong⁶⁰. To ascertain the SSS, we asked the respondents to indicate where on the ladder they stood as a response to the following statement: “Think of this ladder (picture provided) as representing where people stand in your neighbourhood. At the top of the ladder are people who are the best off—those who have the most money, the most education, and the most respected jobs. At the bottom are the people who are the worst off, who have the least money, the least education, and the least respected job or no job. The higher up you are on this ladder, the closer you are to the people at the very top; the lower you are, the closer you are to the people at the very bottom.” There is evidence that the social status of individuals is critical in enhancing resilience to climate-related hazards^{63,64}.

Capturing background characteristics was essential to understanding the context in which respondents and households experienced climate-related hazards, as well as to understanding how these characteristics influenced vulnerability and their ability to adapt. Including these characteristics allowed for a more comprehensive analysis of the relationships between social determinants, environmental risks, and WASH outcomes within informal settlements. Additionally, a thorough description of participants' demographics enables readers and researchers to determine to whom the research findings generalise and facilitates comparisons⁶⁵.

Questionnaire development and data collection

To date, the availability of data collection tools assessing the climate impacts on WASH services at the household level remains limited. Therefore, we adapted the World Health Organisation's (WHO) checklists, which assess flood and drought vulnerabilities in healthcare facilities in the context of climate change, to households in informal settlements⁶⁶. These checklists capture information on the impacts of floods and droughts on WASH services and risk management strategies⁶⁶. Public health, climate change, and WASH experts reviewed the draft questionnaire during stakeholder meetings aimed at developing Uganda's National Health Adaptation Plan⁶⁷, to ensure content and face validity⁶⁸. Upon review, the research team incorporated all the feedback to ensure the relevance and cultural appropriateness of the questions.

Quality control, data management and analysis

Research assistants who completed a five-day training carried out the household interviews. The training provided an overview of the survey and the study tools, as well as the data to be collected. It placed a strong emphasis on ethical considerations in research, particularly the importance of obtaining informed consent, to ensure the integrity of the study. All research assistants had a minimum educational qualification of a Bachelor's Degree in Humanities, Statistics, Environmental Health Science, and Public Health, as well as prior experience in conducting face-to-face interviews. We created the data collection tool using the Kobo Collect online platform to incorporate automated skip logic, pre-coded responses, and mandatory fields, thereby minimising errors, missing data, and inconsistencies. The data management team uploaded the study questionnaire on Android-enabled phones and tablets, which they later used for data collection.

The study questionnaire was originally developed in English and translated into the local languages spoken in the study area (Gishu and Luganda). To ensure translation accuracy and consistency of meaning, the translated versions were independently backtranslated into English by bilingual speakers who were not involved in the initial translation process. The original and back-translated English versions were then compared by the research team to identify and resolve discrepancies through discussion and consensus, ensuring conceptual equivalence across languages. The data collection team subsequently pre-tested both the translated and English versions of the study questionnaire to assess the amount of time required for the interviews and to inform improvements before data collection. Feedback from the pre-testing informed minor revisions to wording and sequencing before final data collection.

The research team conducted the pre-test activity in Katanga slum, an urban informal settlement located in Kawempe division, Kampala City. We selected Katanga as the ideal site due to the similarity in informality^{69–76}. However, we excluded the pre-test results from the main study. The data collection team ensured real-time submission of the data for quality control. Upon submission, the analysts exported the data to the Microsoft Excel Application for thorough cleaning. Data cleaning involved resolving inconsistencies, incomplete responses, and duplicate entries. We performed both descriptive and inferential statistical analyses using STATA version 16.0. Since the impact and risk management scores did not fulfil assumptions for linear regression, i.e. Normality, independence, and constant variance^{77,78}, based on the Shapiro-Wilk and Kolmogorov-Smirnov tests^{79,80}, we applied robust linear regression to model the predictors of flood and drought impacts, as well as risk management scores. Robust linear regression is less sensitive to outliers compared to least squares regression^{81,82}. A p-value of less than 0.005 was considered statistically significant.

Results

Household head background characteristics

The study had 589 respondents. Nearly two-thirds (62.5%) of the households were headed by males. Around two-fifths (40.7%) of the household heads had secondary education, 90.2% were employed, 65.7% rented their homes, and 59.3% lived in permanent structures. Nearly half (44.6%) of the household heads had stayed in the informal settlement for 1–5 years, and close to three-fifths (61.1%) belonged to the low subjective social status (Table 1). The mean subjective social status score was 3.2 (SD ± 1.6) while the median was 3.0 (IQR = 2.0, 4.0). The mean altitude was 1118.6 (SD ± 14.8) while the median was 1118.3 (IQR = 1110.0, 1128.0) metres (Table 1).

Perceived exposure to climate hazards

The perceived exposure of households to climate-related hazards over the past 12 months varied across different types of events. Floods were the most frequently reported hazard, affecting 27.3% of households, followed by water-level rise at 16.1%, drought at 12.9%, and heatwaves at 8.5%. These results highlight that flooding and rising water levels are the most salient and immediate threats to households, particularly in informal settlements where WASH infrastructure and livelihoods are highly vulnerable (Table 2).

Impact of floods on WASH services

About a fifth (22.4%) of the households experienced disruptions to wastewater or sewerage systems, and 21.4% faced increased susceptibility to water-borne diseases due to animal faeces and sewage washing into surface water. Nearly a quarter (23.6%) had a diminished capacity to provide or access safe water for drinking or cooking, and about a fifth (19.5%) were unable to access hygiene services. Around a fifth (22.9%) reported damage to waste storage, leading to environmental contamination, while a similar percentage (22.2%) had reduced ability to maintain waste collection and treatment. Another 22.2% experienced damage to emergency or alternative water sources. Some households (13.6%) had decreased capacity to use toilets or showers, and a slightly higher percentage (16.1%) reported damage to hygiene facilities such as taps. Nearly a quarter (24.3%) faced temporary interruptions in water supply, and some households (14.1%) had reduced capacity to use laundry services, including wire lines. A small fraction (11.5%) reported possible rodent infestation around rubbish bins (Table 2).

Variable	Category	Frequency (N = 589)	Percentage (%)
Sex of household head	Female	221	37.5
	Male	368	62.5
Age	18–24	136	23.1
	25–35	225	38.2
	> 35	228	38.7
Level of education	No formal education	66	11.2
	Primary	238	40.4
	Secondary	240	40.7
	Tertiary	45	7.6
Employment status	Employed	531	90.2
	Unemployed	58	9.8
House ownership	Caretaker	13	2.2
	Family house	71	12.1
	Personal house	118	20.0
	Rented	387	65.7
Type of housing	Permanent	349	59.3
	Semi-permanent	219	37.2
	Temporary	21	3.6
The household has a disabled person	Yes	50	8.5
	No	539	91.5
The household has children below the age of one year	Yes	137	23.3
	No	452	76.7
The household has children below the age of five years	Yes	325	55.2
	No	264	44.8
Duration spent in the study informal settlement (years)	1–5	258	44.6
	6–10	119	20.6
	> 10	201	34.8
Subjective social status	High	19	3.2
	Middle	210	35.7
	Low	360	61.1

Table 1. Household head background characteristics. IQR = InterQuartile Range, SD = Standard Deviation.

Flood impact (N = 589)	%
Experienced disruption of wastewater or sewage systems	22.4
Experienced contamination of the primary water source due to damage to the sewage system	19.0
Experienced increased susceptibility to water-borne diseases as animal faeces and sewage got washed into surface water	21.4
Experienced reduced capacity to provide or access safe water for drinking or cooking	23.6
Unable to access or provide hygiene services	19.5
Suffered environmental contamination by biological and chemical hazards owing to damage to waste storage systems	22.9
Suffered reduced capacity to maintain waste collection and treatment	22.2
Suffered damage to emergency or alternative water sources	22.1
Reduced capacity to use toilets or showers	13.6
Suffered damage to hygiene facilities such as tippy taps	16.1
Suffered a temporary water supply interruption	24.3
Experienced reduced capacity to use laundry services	14.1
Experienced rodent infestation around rubbish bins due to floods	11.5

Table 2. Impact of floods on household-level WASH services in informal settlements in Mbale city, Uganda. Values indicate the percentage of households experiencing each flood-related impact.

Risk management for floods

The households had few preventive measures in place for addressing water and sanitation challenges. Only a small fraction (2.4%) had had a natural floodwater infiltration system, and a slightly higher (5.3%) had anti-mosquito breeding measures. Less than a tenth (9.0%) had a schedule for emptying latrines before the flood season. A few households (6.3%) had a safe waste disposal system before, during and after floods, and even a

Flood risk management measures (N=589)	%
With a natural floodwater infiltration system to reduce the risk of flooding	2.4
With anti-mosquito breeding measures	5.3
With a schedule for emptying latrines in advance of the flood season to avoid overflows	1.5
With a safe waste storage place that is uninterrupted by floods	6.3
With a safe waste disposal system before, during and after floods	3.9
With an established safe management approach to transport waste in case of floods	2.7
With water storage tanks appropriately covered to prevent access or contamination, and safety measures in place for flooding events	1.0
With on-site water purification equipment to provide safe drinking water	2.7
With waste pits able to withstand flood events	2.0
Kept waste sealed in rubbish bins to avoid rodents	3.9

Table 3. Flood risk management among households in informal settlements, Mbale city, Uganda. Values indicate the percentage of households with each flood risk management strategy.

Variable	Flooding impact scores		Flood risk mitigation	
	β coefficients (95% CI)	p-value	β coefficients (95% CI)	p-value
Altitude (meters above sea level)	- 0.047 (- 0.078, - 0.016)	0.003	0.001 (- 0.004, 0.005)	0.982
Sex of household head				
Female	1		1	
Male	0.396 (- 0.337, 1.128)	0.289	- 0.082 (- 0.252, 0.086)	0.336
Employment status				
Employed	1		1	
Unemployed	- 0.394 (- 1.558, 0.769)	0.506	- 0.086 (- 0.313, 0.141)	0.456
Housing type				
Permanent	1		1	
Semi-permanent	- 0.038 (- 0.775, 0.699)	0.919	- 0.106 (- 0.261, 0.049)	0.180
Temporary	4.276 (1.935, 6.618)	<0.001	0.171 (- 0.188, 0.531)	0.349
Subjective social status score	- 0.296 (- 0.551, - 0.040)	0.024	0.009 (- 0.053, 0.708)	0.771
Have children under five	- 0.648 (- 1.361, 0.064)	0.075	- 0.002 (- 0.161, 0.157)	0.982
Have a disabled person	1.241 (0.041, 2.442)	0.043	0.021 (- 0.212, 0.255)	0.857

Table 4. Unadjusted models for the predictors of flooding impacts and flood risk mitigation scores.

smaller percentage (2.7%) had an established safe management approach to transport waste in case of floods. Only one per cent had water storage tanks appropriately covered and located for flooding events, and a similar percentage (2.7%) had on-site water purification equipment. A few households (2.0%) had waste pits able to withstand flood events, and some (3.9%) kept waste sealed in rubbish bins to avoid rodents (Table 3).

Predictors of flooding impacts and flood risk mitigation scores

Area altitude, temporary housing, subjective social status, and having a disabled person significantly affected flooding impact scores. Each additional meter of altitude reduces the impact score by 0.047 ($p=0.003$), indicating lower flood severity at higher elevations. Temporary housing increases the impact score by 4.276 points ($p<0.001$) compared to permanent housing, reflecting greater vulnerability due to less robust structures. Higher subjective social status scores seem to lower the impact score by 0.296 ($p=0.024$), suggesting better resources or safer locations, while households with a disabled person face a 1.241-unit higher impact score ($p=0.043$) due to increased vulnerability. No variables significantly predict flood risk mitigation, indicating these factors do not strongly influence mitigation behaviours (Table 4).

Impact of drought on WASH services

The households also faced challenges related to water scarcity and quality. About a tenth (11.4%) lacked water for washing, cooking, and hygiene, and 11.0% had low water quality. Some households (8.1%) experienced increased water salinity in their groundwater resources, and a slightly higher percentage (9.5%) had no access to potable water for drinking and cooking purposes. A few households (4.7%) reported an increased rate of broken pipes. Only a small fraction (7.1%) had reduced function of sanitation systems and hygiene practices, and some (8.5%) had increased dependence on less secure alternative water sources. A few households (4.9%) had a possible increase in vector breeding sites due to inadequate water storage in the household or surrounding areas (Table 5).

Drought impacts (N=589)	%
Had water for washing, cooking and hygiene	11.4
Experienced increased water salinity in groundwater resources due to decreased recharge	8.1
Had access to potable water for drinking and cooking	9.5
Had low water quality	11.0
Had an increased rate of broken pipes	4.7
Had reduced function of sanitation systems and hygiene practices (flush toilets, showers, sewerage, hand washing)	7.1
Had increased dependence on less secure alternative water sources	8.5
Had a possible increase in vector breeding sites due to inadequate water storage in the household or surrounding areas	4.9

Table 5. Impact of drought on WASH services among households in informal settlements in Mbale city, Uganda. **Values indicate the percentage of households (HH) experiencing each impact.

Drought risk management measures (N= 589)	%
Had a water management system to avoid or reduce vector breeding sites	6.3
Had anti-mosquito breeding measures to avoid vector-borne diseases	6.1
Had a rainwater catchment system with safe water storage	5.4
Had water storage tanks with appropriate covers to prevent contamination	2.4
Had water storage that is protected from direct sunlight	4.2
Were able to provide sufficient drinking water for visitors	10.5

Table 6. Risk management for droughts among households in informal settlements in Mbale city, Uganda. Values indicate the percentage of households with each drought risk management strategy.

Risk management strategies for droughts in informal settlements

The households had limited water management and protection measures. Only a few households (6.3%) had a water management system to avoid or reduce vector breeding sites, and a similar percentage (6.1%) had anti- mosquito breeding measures. A slightly lower percentage (5.4%) had a rainwater catchment system with safe water storage. Only a small fraction (2.4%) had water storage tanks with appropriate covers, and some (4.2%) had water storage that was protected from direct sunlight. About a tenth (10.5%) had sufficient drinking water for visitors (Table 6).

Predictors of drought impact and drought risk mitigation scores

Temporary housing and subjective social status significantly influenced drought impact scores, with temporary housing linked to a 0.656-unit higher score ($p < 0.001$) compared to permanent housing, and higher social status reducing the score by 0.087 ($p = 0.048$), reflecting better resilience. For drought risk mitigation, semi-permanent and temporary housing reduce mitigation scores by 0.201 ($p = 0.010$) and 0.391 ($p < 0.001$), respectively, compared to permanent housing, suggesting that less stable housing types are associated with less effective drought mitigation strategies, likely due to limited access to resources or infrastructure (Table 7).

Discussion

This study analysed how climate-related hazards - drought and floods impact WASH and the attainment of other SDGs. We begin by illustrating how the impacts of different droughts and floods can affect the attainment of other sustainable development goals, followed by an examination of the relationships between household characteristics and flood and drought impacts, as well as risk management scores.

Association between household characteristics and flood and drought impacts and risk mitigation

This study found that low area altitude, i.e., the height of a location above sea level⁸³, significantly increased flood impacts within informal settlements. Specifically, individuals who inhabited areas of low altitude were more likely to face more severe impacts compared to those who did not. A lower altitude is associated with a proximal vertical distance to water levels, while a higher altitude is associated with a distant vertical distance from them. Our findings are consistent with previous literature^{84,85}, which indicated that lower topographies are associated with an increase in flood disasters, and the fact that low-altitude areas have a higher likelihood of flooding. Individuals at lower altitudes were therefore more prone to intense flooding and its associated consequences, including disruption of wastewater or sewage systems, increased susceptibility to water-borne diseases, and an inability to provide hygiene services. In Mbale, households located in low-lying areas are also more prone to water accumulation and prolonged flooding due to poor drainage and runoff convergence, increasing the risk of contamination of water sources, damage to sanitation facilities, and disruption of waste management services⁸⁶. Conversely, households situated on steep or elevated slopes, particularly along the foothills of Mount Elgon, may experience increased exposure to rapid runoff, soil erosion, and landslides, which can damage housing structures, latrines, and water infrastructure⁸⁷. These elevation-related differences affect both the frequency

Variables	Drought impact scores		Drought risk mitigation	
	β coefficients (95% CI)	<i>p</i> -value	β coefficients (95% CI)	<i>p</i> -value
Altitude (meters above sea level)	−0.006 (−0.015, 0.003)	0.214	−0.001 (−0.006, 0.003)	0.617
Sex of household head				
Female	1		1	
Male	−0.209 (−0.520, 0.101)	0.186	−0.085 (−0.250, 0.081)	0.315
Employment status				
Employed	1		1	
Unemployed	−0.092 (−0.534, 0.350)	0.683	−0.120 (−0.362, −0.122)	0.330
Housing type				
Permanent	1		1	
Semi-permanent	0.052 (−0.261, 0.364)	0.746	−0.201 (−0.354, −0.048)	0.010
Temporary	−0.656 (−0.852, −0.460)	<0.001	−0.391 (−0.543, −0.237)	<0.001
Subjective social status score	−0.087 (−0.172, −0.001)	0.048	−0.011 (−0.067, 0.046)	0.706
Have children under five	−0.010 (−0.392, 0.198)	0.519	−0.023 (−0.185, 0.139)	0.782
Have a disabled person	0.252 (−0.274, 0.778)	0.348	−0.013 (−0.260, 0.234)	0.916

Table 7. Unadjusted models for the predictors of drought impact and drought risk mitigation scores.

and severity of flood impacts, thereby influencing household-level vulnerability to climate-related hazards. To mitigate the impact of low topographies, house elevation is recommended to increase the vertical distance from the flood level^{88,89}. Elevating houses refers to locally observed flood-adaptation practices aimed at reducing water entry into living spaces. These practices include raising house foundations using soil, stones, or compacted earth⁹⁰.

Households living in temporary houses, particularly those made of mud and wattle and roofed with tarpaulin or grass, reported higher flood impact scores. These houses, being highly permeable, allow for easy flood penetration. Unlike permanent housing, which is typically constructed from impermeable materials like brick and concrete, mud and wattle (from which temporary houses are made) can rapidly absorb water, thereby increasing flood impacts. In addition to residing in temporary housing, individuals with low subjective social status scores experienced higher flood impact scores. Having higher scores is associated with better financial capacity, education and respected jobs. Access to finances enables households to implement flood risk mitigation measures. Education is critical in informing risk management and adaptation as it reshapes behaviours, develops skills, and spurs innovation. Having a respected job instils a sense of responsibility, which is crucial for residing in safer zones. Previous literature has shown that the education and income of the household head contribute to building climate resilience^{10,91–93}. These findings demonstrate how investments in education and financial literacy can mitigate the impacts of floods and stimulate innovation in risk management.

Implications of droughts and floods impacts on other sustainable development goals

In Mbale, recurrent flooding had direct implications for progress towards several SDGs, particularly at the household and community level. In this study, over 10% of households lost access to safe water and sanitation. Floods damage water infrastructure, impeding the availability, quality, and accessibility of safe water sources^{94–96}. In Eastern Uganda, where Mbale is located, floods were reported to damage homes, contaminate water and increase exposure to waterborne diseases, which compromises the attainment of SDGs 3 and 6^{97,98}. The presence of high levels of faecal contamination, other pollutants, and debris in floods contaminates water, making it unsafe for consumption unless treated^{99,100}. Additionally, 22.2% of the households reported reduced capacity to manage waste collection and treatment due to the floods. These disruptions compromise SDG 6 (Clean Water and Sanitation), particularly target 6.2 on safe disposal and hygiene services.

Flood-induced damage to waste storage systems including the overflow and dispersion of waste due was experienced and this can lead to the release of waste materials, including hazardous substances, into the surrounding environment^{101,102}. SD6 11 emphasises air quality and waste management. Damaged containment undermines the efforts of proper waste management (SDG 11.6.1), thus contributing to pollution of land (SDG 15), air (SDGs 3.9, 11.6 & 13) and water (SDG 14.1).

Additionally, floods damaged sanitation infrastructure, rendering some facilities unusable. The lack of functioning toilets and handwashing facilities increases the risk of waterborne diseases and the spread of infections within affected communities. Access to safe water is crucial for achieving SDG 6 target 6.2. It enables the safe disposal and treatment of excreta and the provision of a functional hand hygiene service. It also contributes to good health and well-being (SDG 3), particularly through a reduction in maternal and under-five mortality (targets 3.1 and 3.2), as well as waterborne diseases (target 3.3). A lack of adequate water can hinder hygiene practices and increase the risk of infections, such as maternal and neonatal sepsis^{103,104}. Moreover, diarrhoea, which is a significant cause of under-five mortality, can be exacerbated by limited access to safe water for drinking and hygiene^{105,106}.

Limited access to safe water for food hygiene has also been reported to contribute to diarrhoea. Consequently, it hinders the attainment of SDG 2 targets, which include reducing undernourishment, food insecurity, stunting, malnutrition, and anaemia. Floods and dry spells are the most significant risks reported by smallholder

horticultural farmers in Mbale, causing major yield losses and food shortages, pushing households deeper into poverty^{11,107}. The relationship between limited access to safe water, malnutrition, stunting and anaemia is widely documented in the literature^{108,109}. WASH-related diseases are also known to affect school attendance and educational outcomes (SDG 4)^{110,111}. Floods and landslides in Mbale have led to temporary school closures and hindered educational attainment¹⁰⁷.

Compromised health and well-being (SDG 3), limited access to quality education, and an increase in malnutrition (SDG 2) lead to increased expenditure on healthcare, thereby compromising the ability of households to eliminate poverty (SDG 1) (vicious cycle of poverty). In Mbale, marginalized communities report declining resilience and increased health risks during extreme climate-related events¹⁰⁷. The destruction of water and sanitation infrastructure compels victims to seek alternatives that do not offer the same level of safety and security. Thus, some household members may resort to open defecation or use unimproved water sources or shared sanitation facilities. Also, inadequate water and sanitation infrastructure predisposes women and girls to violence and compromises menstrual hygiene through reduced privacy. This limits progress toward achieving SDG 5. Droughts have significant impacts on water availability and can lead to increased conflict among competing users. The Karamojong and Turkana conflict in Eastern Africa is an example of how droughts can hinder peace, inclusiveness, and justice¹¹². Droughts reduce the amount of water available for household chores and drinking, increase the salinity of groundwater resources, and force reliance on less secure alternative water sources¹¹². Limited availability of water resources can exacerbate conflicts between different user groups. It is important to analyse the impact of droughts on water resources, as well as the role of climate variability and water governance in exacerbating water scarcity¹¹³. The consequences of droughts on water availability can have far-reaching implications for human life and safety¹¹⁴. Therefore, it is crucial to prepare and maintain adequate water reserves to mitigate water shortages.

Approximately one-third of the households reported damage to emergency or alternative water sources. This affects the sustainable withdrawal of freshwater, leading to water stress (SDG 6.4.2). Available evidence also indicates that damaged water infrastructure contributes to a drop in water service levels at both household and institutional levels^{95,115}. For example, households or institutions that could have attained a better water service level may be affected due to the impacts of climate hazards on water supply. The damage to water supply systems increases non-revenue water, resulting in economic loss (SDG 8), and hinders an effective response to humanitarian crises (SDG 16). The damage to water sources in Mozambique¹¹⁶ clearly illustrates how humanitarian response is affected (SDG 16).

The disruption of wastewater or sewerage systems resulted in cross-contamination from damaged sewer pipes, and increased susceptibility to water-borne diseases due to animal faeces and sewage being washed into surface water. Climate-related hazards, such as floods, compromise the ability to achieve SDG 14 (14.1 and 14.3) by increasing pollution of water sources, acidification, and eutrophication. Floods increase nutrient loading and consequently eutrophication, which is reported to be detrimental to some aquatic life¹¹⁷. Disruption of wastewater and damage to sewer pipes due to floods contribute to the introduction of invasive alien species in land and water ecosystems (SDG 15). Additionally, the disruption of wastewater or sewerage systems can affect the economy by causing business closures, limiting access to a decent workplace, and loss of productivity (SDG 8.2), as reported by Hammond et al.¹¹⁸. Faecal contamination compromises food safety (SDG 2.1), leads to food losses since some foods are rendered unfit for consumption (SDG 12), and loss of access to markets (SDGs 8, 9, and 17). Floods lead to the disruption of wastewater systems, which increases the amount of waste (liquid and solid) generated (SDG 12.5). This poses challenges in the management of huge volumes of waste, as large volumes of storm water mixed with sewage overload sewage treatment plants (SDG 11.6.1). The impact of climate-related hazards on waste and sewerage systems has also been reported by other scholars^{21,119,120}.

Floods led to damage to waste storage facilities in nearly a quarter of the households, thereby causing environmental pollution by biological and chemical hazards. Additionally, 22.2% of the households had reduced capacity to maintain waste collection and treatment as a result of floods. Flood-induced damage to waste storage systems, as well as the overflow and dispersion of waste due to flooding, leads to the release of waste materials, including hazardous substances, into the surrounding environment^{101,102}. SDG 11 emphasises air quality and waste management. Damaged containment undermines the efforts of proper waste management (SDG 11.6.1), thus contributing to pollution of land (SDG 15), air (SDGs 3.9, 11.6 & 13) and water (SDG 14.1).

Damage to waste storage exposes waste to environmental conditions that lead to the putrefaction of organic matter and creates conducive habitats for vector and vermin breeding. Poor waste management is also reported to contribute to the emission of greenhouse gases, including methane (SDG 13.2.2 and SDG 11.6.1)^{101,102}. The interconnectedness between waste management and climate change highlights the need for a nexus approach in addressing global environmental change. More than a tenth of the households had reduced capacity to maintain waste collection and treatment. Climate hazards disrupt the regular operations of waste management systems, leading to accumulated waste in households and communities. Climate hazards such as floods hinder the sustainability of cities and communities (SDG 11) by undermining waste management systems and the right to a clean urban environment (SDG 11). Furthermore, it also impacts responsible consumption and production by impeding efforts to promote circular waste management practices (SDG 12).

Strengths and limitations

This study adapted the WHO's checklist for assessing vulnerabilities in healthcare facilities in the context of climate change to households, which provides a new dimension for assessing vulnerabilities in households. The study utilises a large sample size, instilling confidence in the insights into climate impacts and risk mitigation in informal settlements in other developing economies. Despite these strengths, the study was prone to recall bias. For some households, a considerable amount of time had passed between exposure to climate impacts and the data collection period, making the recall of the impacts challenging. In anticipation of recall bias, we trained

research assistants on using event calendars and probing to mitigate the risk, and also used a short recall period of 12 months.

The impact scoring method used assumed equal contribution of all indicators and does not capture differences in severity, duration, or magnitude of individual impacts. The scoring reflects the presence of reported disruptions rather than their intensity, which may mask variations in lived experiences. Future studies could incorporate severity-based weighting or longitudinal measures where such data are available. Despite these limitations, the method provides a practical approach for assessing flood-related WASH impacts in data-constrained urban settings.

Conclusions

The study assessed the household vulnerability to climate impacts on WASH and risk mitigation in informal settlements of Eastern Uganda. Household characteristics such as temporary housing, lower subjective social status, area altitude, and the presence of a person with a disability significantly influence exposure to flood and drought impacts, as well as households' capacity to manage risks. Floods were found to damage WASH infrastructure, contaminate water sources, and increase the risk of waterborne diseases, while droughts reduced water availability, increased salinity, and heightened competition for scarce resources. These climate impacts have cascading effects on health, education, food security, gender equality, poverty reduction, and urban resilience. Strengthening partnerships, applying nexus approaches, and improving water governance are critical for enhancing household and community resilience to climate hazards and safeguarding sustainable WASH services.

Data availability

The datasets analysed in this manuscript are available from the corresponding author upon reasonable request.

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Author contributions

JB, TS, and RKM obtained the funding for this study. JB, TS, PT, RKM, RO, MMM, TA, DAY, STW, and DS conceptualised the study, participated in data collection and analysis, and drafted the manuscript. JM, PT, DN, CHM, JG, DN, SM, RN, MN, AN, BNT, and AG participated in the analysis and drafting of the manuscript. All authors read and approved this manuscript before submission to this journal.

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Declarations

Ethics approval and consent to participate

This study was conducted following the Declaration of Helsinki. Ethical approval was obtained from the Makerere University School of Public Health Research and Ethics Committee (Protocol SPH-2023-464) and was registered by the Uganda National Council for Science and Technology (HS3168ES). Before data collection, the investigators sought administrative clearance from the Mbale City Council authorities and the leadership of the respective informal settlements. All participating households provided informed written consent.

Competing interests

The authors declare no competing interests.

Additional information

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